Introduction

Global health concerns regarding flaviviruses continue to grow as anthropogenic activities and ecological changes expand the geographic range of their vectors, exposing previously unaffected populations. Such concerns with Flaviviruses, a group of RNA viruses infecting over 400 million people annually (Pierson, T.C., 2020), is their transmission by mosquitoes and ticks. Such vectors make regions experiencing outbreaks much harder to mitigate. Within this group, West Nile Virus (WNV) can cause severe symptoms, such as encephalitis and meningitis, in elderly or immunocompromised individuals.

West Nile Virus was selected as the prototype virus due to its close genetic relationship with other flaviviruses, allowing insights gained to potentially contribute to further research on treatments for a broader range of viruses within this group. A flavivirus for which no effective human vaccine currently exists. It is essential to note that the CDC reports that of the arboviral disease cases reported, 90% were neuroinvasive and WNV was found to be responsible for 96% of the neuroinvasive cases (Fagre, A.C., 2023).

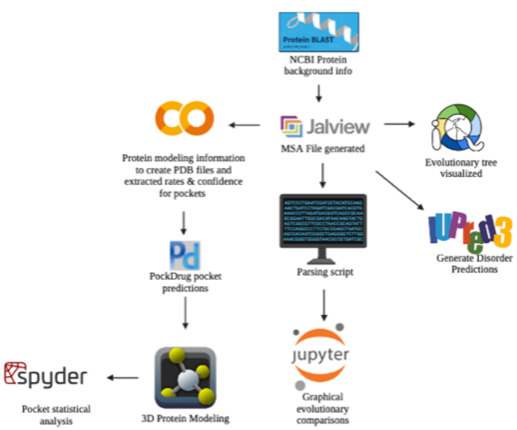
With current treatment, there exist issues with vaccines for certain Flavivirus. The issue, known as antibody-dependent enhancement (ADE), is a cross-reactivity immune system response linked to some vaccine treatments and poses a significant challenge for flavivirus management. For this reason, this research focuses on the development of antiviral drugs.

The study targets the non-structural protein 5 (NS5), a key protein involved in viral replication and RNA capping. Disrupting NS5 function has the potential to significantly reduce infection rates and disease severity, making it an ideal candidate for antiviral drug development.

Objective

This study aims to present viable drug candidates for West Nile Virus and other viruses in the Flavivirus family, using a bioinformatics approach.

Methods



Results

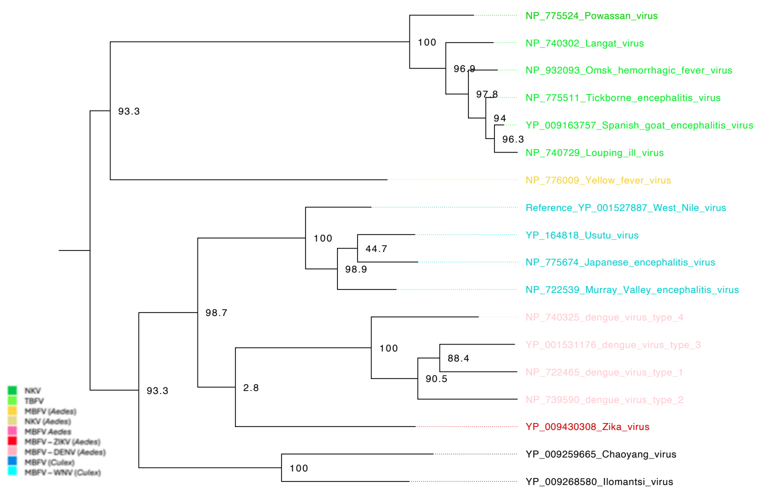


Figure 1. Phylogenetic tree displaying the relationship between ref. WNV and other Flavivirus

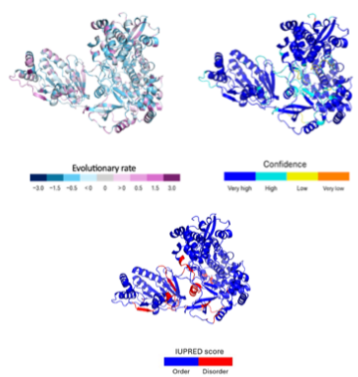


Figure 2. Protein NS5 3D visualization colored by evolutionary rate, model confidence, and IUPRED.

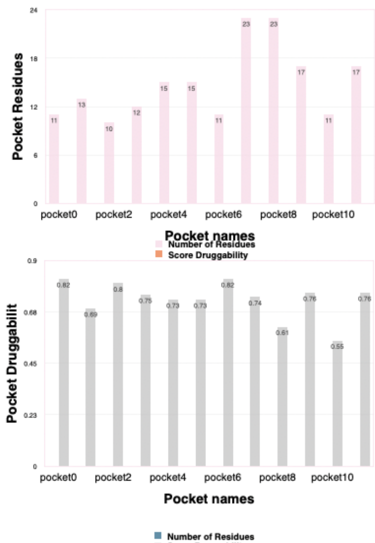


Figure 3. Protein pockets residue numbers and druggability percentage visualized.

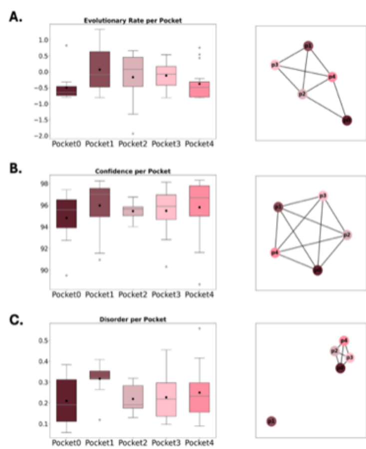


Figure 4. Pocket comparison visualizations via boxplots and networks.

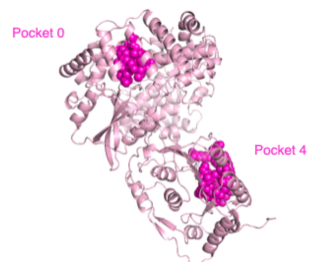


Figure 5. Protein NS5 in light pink and pocket 0 (org. 29) and pocket 4 (org. 8) visualized through 3D modeling in PyMOL.

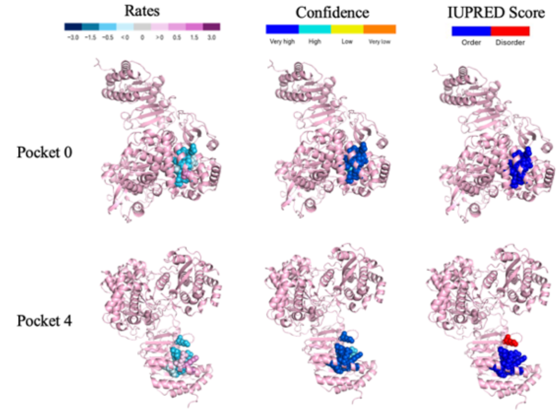


Figure 6. Protein NS5 in light pink and pocket 0 (org. 29) and pocket 4 (org. 8) visualized through 3D modeling in PyMOL.

Discussion

The results of this research found West Nile Virus NS5 to be a large protein with many useable pockets. The NS5 protein itself is predicted with low evolutionary rates, high modeling confidence, and low disorder percentage. This further highlights the protein as a viable drug target as these characteristics improve the possibility of drug efficacy for a longer duration- standing against evolving proteins. While also understanding targeting it would lead to affecting viral replication and capping.

Further analysis with bioinformatics tools and comparisons revealed **pocket 0 and pocket 4 to be the most promising drug targets** when compared to other pockets in the protein. Pockets 0 and 4 both have low evolutionary rates, high modeling confidence, and low disorder percentages. Furthermore, both of these pockets can be easily accessed and are not found deep within the protein. These key characteristics mean a drug created will have a higher possibility of being effective after being modeled to a structure that is more confidently reported theoretically. Also, a low evolutionary rate means the drug can retain efficacy for longer as the protein will not evolve protections against it quickly. This study's limitations lie in its theoretical approach- it was all done by computer programming. The next step would be empirical research testing antiviral activity in cell-based assays or in vitro systems using West Nile virus cultures.

References

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Acknowledgements

This project was completed as part of a Course-based Undergraduate Research Internship Experience (CURIE). Research reported was supported by the National Institutes of Health under award number R25AI170382. The content is solely the responsibility of the authors and does not necessarily represent the views of the NIH.